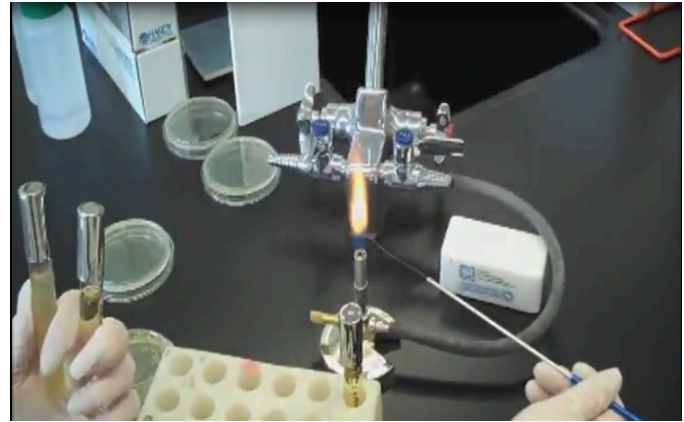


Case Study

Jeanne is a laboratory technician at BactoLabs, a clinical microbiology testing facility. Her job involves culturing and testing patient specimens for the identification of infectious microbes.

One week ago, Jeanne received a sample from a suspected case of bacterial meningitis. She took steps to culture and test for the suspected causative agent, *N. meningitidis*, per standard protocol.

The tests conclusively demonstrated that the patient did in fact have meningococcal meningitis.

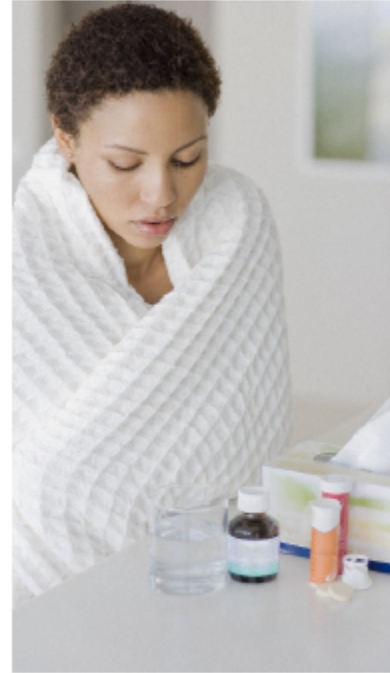


Infection Acquired in the Laboratory

Five days later, Jeanne was admitted to the hospital with a severe headache, vomiting, fever and confusion.

At the hospital, Jeanne was tested and diagnosed with meningococcal meningitis.

Later in this lab exercise, we will revisit Jeanne's case to determine what might have occurred.



Sam Edwards/age fotostock

Ubiquity of Bacteria

Bacteria are ubiquitous, referring to the fact that they are found nearly everywhere.

Bacteria are abundant in soil, water, and the Earth's crust. If sampled, most environments will show some level of bacterial growth. In fact, even deep ocean vents and geysers have microbial life.

Bacteria also live in and on our bodies, actually outnumbering our own human body cells approximately 10 to 1! Most of the time, these normal bacteria use humans as hosts without causing any disease, but there are also many bacteria that are disease-causing parasites.



Places Without Bacteria

There are plenty of places where you won't normally find bacteria, such as the inside of properly canned food.

More obvious places where you should not find bacteria are sterile fields in an operating room or sterile environments in a lab.

In most cases, however, special techniques have been used to remove or destroy normally existing bacteria in these environments.



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Factors that Influence Microbial Growth

The growth and survival of microorganisms is greatly affected by a variety of environmental factors. Microorganisms have to adapt to whatever variations they encounter in order to survive.

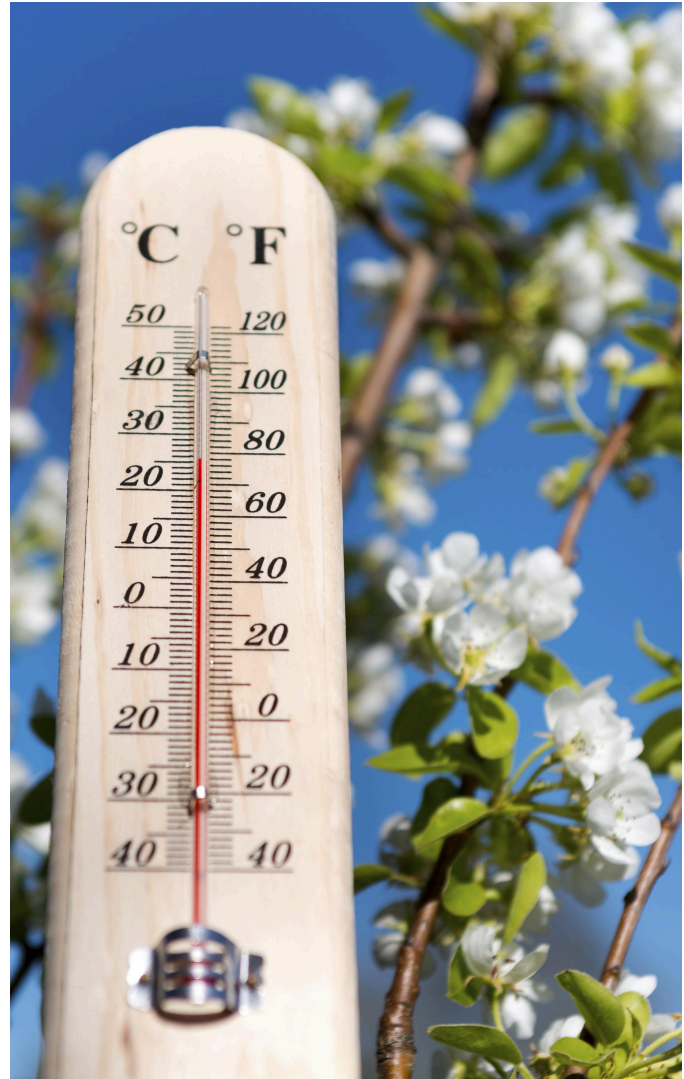
Environmental factors that influence microbial growth include temperature, osmotic pressure, atmospheric gases, pH, and moisture.

Although extreme microbes exist in even the harshest habitats, most organisms prefer moderate conditions.

For example, most microbes prefer warmer temperatures in the range of 20 to 40 degrees Celsius. Environments with moisture are more conducive to microbial growth than extremely dry habitats.

Most microorganisms are neutrophiles, meaning they prefer environments with a pH near to neutral. And finally, a large majority of microbes either require or are capable of growth in an oxygen-rich environment.

Therefore, when you sample different environments, you are likely to see varying levels of microbial contamination.



Kristian Husar/Getty Images



Javier Larrea/age Fotostock

Correct Labeling of an Agar Plate

It is very important that you always label your plate correctly. Plates are labeled on the bottom, just in case the lid would become separated from the actual culture.

This also makes it easier to read the labels when the plates are on the incubator shelves, as all plates are placed in the incubator lid side down.



Position for Incubating Agar Plates

During incubation, agar plates should be placed into the incubator with the lid down.

This prevents condensation from forming on the lid and dripping onto the bacterial culture.

If condensation did form, it could drip onto the bacterial culture and cause contamination or dispersal of the colonies.



Typical Incubation Temperature

The majority of bacteria are mesophiles, meaning that they prefer temperatures in the range of 20-40° Celsius.

Human body temperature is 37° Celsius, a temperature at which many environmental bacteria also thrive.

Therefore, for most lab exercises, incubators are set to 37° Celsius.

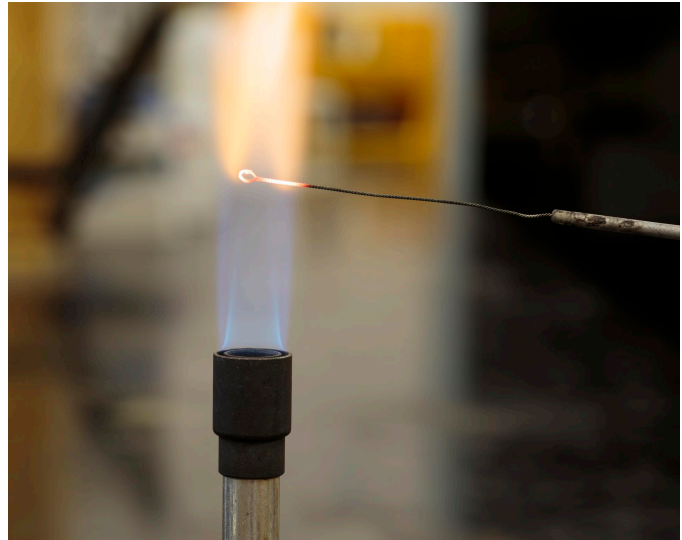


Pixtal/AGE Fotostock

Goal of Aseptic Technique

The goal of aseptic technique is to prevent the contamination of yourself and your microbial cultures when working with microorganisms.

Therefore, you should take various steps during microbiology laboratory exercises to sterilize or disinfect materials.



Geoffrey Whiteway/123RF

Pure Cultures

In many cases, pure cultures are necessary if we are trying to identify one particular organism.

Pure cultures are those that contain only one type of organism.

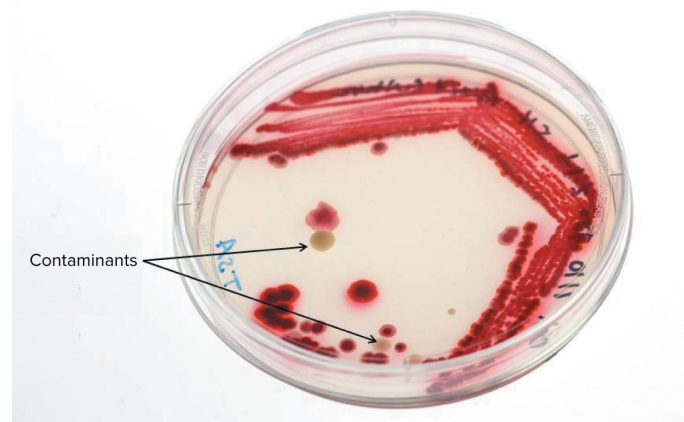
Aseptic technique prevents the introduction of unwanted microbes or contamination of a culture. Therefore, aseptic technique is an important part of making pure cultures.



Contaminated Cultures

Contaminated cultures contain unwanted organisms.

If correct aseptic techniques are not used, it is possible that your culture could be contaminated with organisms from your hands, the environment, or the tools you are working with.



Disinfection of the Lab Bench

It is important that the laboratory work area is treated with a disinfectant prior to and after working with cultures.

Remember that the disinfection process destroys most microorganisms and viruses, and it reduces the risk of contamination to cultures and the laboratory worker.



Culture Media

Broth



Agar Slant



Agar Plate



Like any other living organisms, microbes need the right kind of nutrients to grow and reproduce.

The use of **culture media** with specific compositions and consistency helps you grow microbes in a controlled manner.

Media such as broths and milks come in a liquid form. Some media have agar added to give them a solid consistency. These solid media are prepared in Petri plates or as slants in culture tubes.

Knowing the different ways in which culture media is handled will help you produce cultures without contaminating them with unwanted microbes.

Inoculating Tools

Microbial cultures are transferred from one medium to another by inoculating loops or needles.

The use of either the loop or the needle is dependent on the specific type of transfer you are conducting.

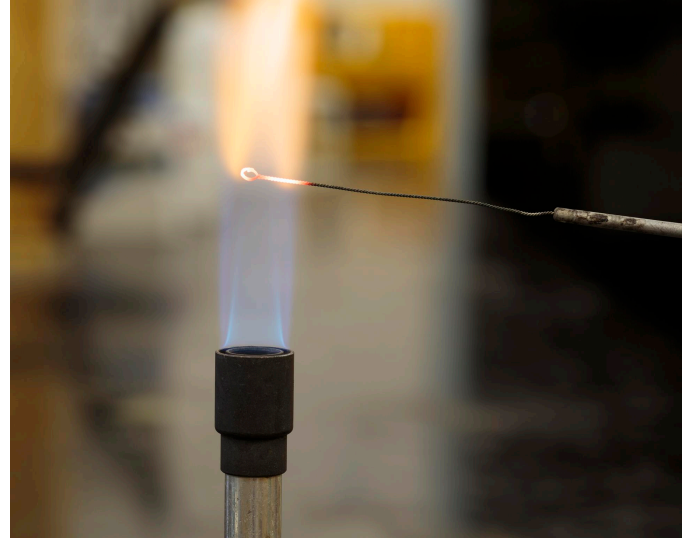
Loops are usually best when transferring an inoculum between broths or onto the surface of slants and plates. Needles are used to stab an agar slant from the top surface to the bottom of the tube.



Sterilization of the Inoculating Tools

It is crucial that the inoculating tool is sterilized **before** and **after** transfer, and this is usually accomplished by using dry heat from a Bunsen burner or an incinerator.

Then, the tool must be cooled before picking up the cells to be transferred to ensure that they are not destroyed.



Geoffrey Whiteway/123RF

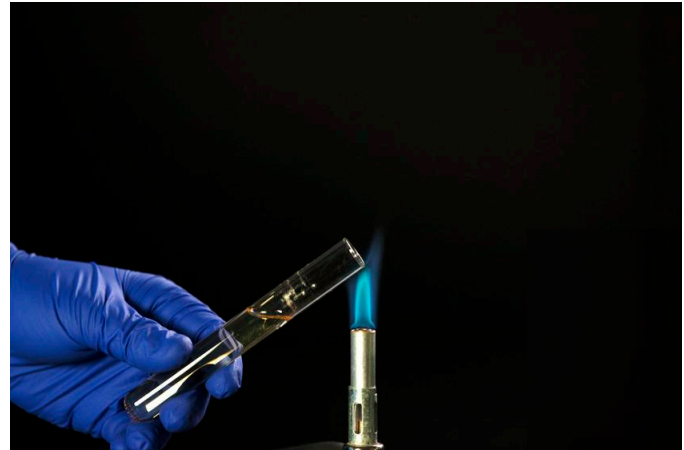
Working with Culture Tubes and Agar Plates

Careful handling of cultures and sterile medium containers also helps to prevent contamination.

The mouth of a glass culture tube should be passed through the flame of a Bunsen burner when the cap is removed and just before the cap is replaced.

The cap of the tube should never be placed on the bench, but instead, it should be held by the little finger of one hand.

For agar plates, it is good practice to use a technique called “clamshelling”. For this, you simply raise only one side of the plate lid off of the plate while removing or adding bacteria to the surface.



Somprasong Wittayanupakorn/123RF

What Is a broth

A broth is a liquid nutrient medium used to support the growth of microorganisms.

The advantage of a broth medium is the ability to support rapid and large bacterial population growth.

Patterns of growth in broth tubes can be used in the bacterial identification process. Broths are also commonly used for antimicrobial sensitivity assays.

Limitations do exist, however. In a broth, you cannot see individual bacterial cells or colonies. Therefore, you cannot separate species of bacteria from a broth culture to create pure cultures.



What Is a slant?

An agar slant is a culture tube with a medium that has solidified with an angled surface. The slanted surface provides a greater surface area for bacterial growth.

When inoculating slants, bacteria can be spread on the slanted surface by either a loop or a needle. A needle can also be used to stab an inoculum from top to bottom of the medium in the tube.

Many biochemical tests involve the use of specialized media in slants. This type of medium is also an excellent choice for storage of stock bacterial cultures.



Transfer from a Slant Culture to a Sterile Slant

Here are the correct steps for an aseptic transfer of bacteria from a slant culture to a sterile slant.

Heat the inoculating loop until it is red-hot. Remove the cap of the slant culture and flame the mouth of the open tube.

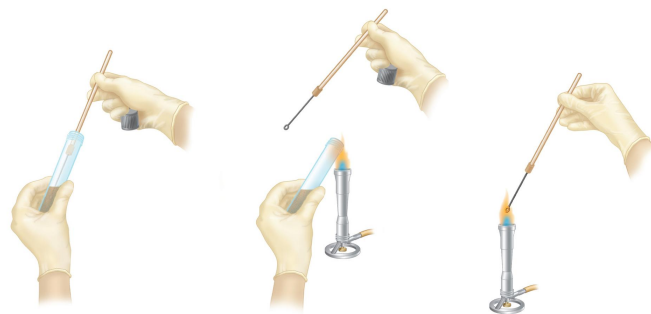
Take a small sample of the organisms using the loop. Flame the mouth of the culture tube again, and replace the cap. The slant culture tube can be placed back in the rack.



Transfer from a Slant Culture to a Sterile Slant

Remove the cap from the sterile slant and flame the mouth of the tube. Streak the surface of the slant in either a zigzag or straight line manner.

Flame the mouth of the tube again and replace the cap. Heat the loop again until it is red-hot.



What Is an Agar Plate?

Agar plates are Petri dishes with solidified culture medium. Plates provide a large surface area for microbial growth and are often used to isolate individual bacterial colonies.

Colony morphology can be observed as an initial step in identifying bacteria. Sometimes, specialized plate media are used to further aid in bacterial identification.

